

Two *NR4A3* fusion-junction gapmers for extraskeletal myxoid chondrosarcoma, screened against wild-type parents

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Abstract

Extraskeletal myxoid chondrosarcoma (EMC) is an ultra-rare sarcoma defined in most cases by an in-frame fusion of a variable partner gene to *NR4A3*. That junction is present in no normal transcript, so an antisense gapmer could in principle cleave the fusion and spare both parents. This work is computational: nothing was synthesised or tested, and every sequence named is a research reagent not for administration. Of 190 junction-spanning 16-mers tiled at 5-6-5 across the 38 in-frame junctions of five modelled partners, 87 let a mature wild-type parent transcript pair their whole catalytic gap over ten or more contiguous base pairs, 61 of those against wild-type *NR4A3* itself. Two reagents are named at the two most frequently reported breakpoints: 5'-GGGCATATCATCAAAC-3' at *EWSR1* exon 12 and 5'-GGGCATATCTTGTGTG-3' at *TAF15* exon 6, both at the panel's top gap-level margin of three, with longest wild-type parent runs of eight and nine base pairs. Five test articles are named: three engineered constructs, two of them at these junctions, and two fusion-positive EMC cell models whose reported *NR4A3* exon-2 acceptors are matched to different designs. The design pipeline is released; its published-breakpoint list waives seams that would otherwise be refused rather than gating which junctions it will design for.

1 • Background

EMC is defined in the large majority of cases by an in-frame fusion of *EWSR1* to *NR4A3*,¹ with *TAF15* a substantial minority and *TCF12* and *TFG* rare.² The disease responds poorly to conventional cytotoxic chemotherapy,³ though responses do occur: one molecularly confirmed series recorded four partial responses in ten evaluable patients.⁴ A tyrosine-kinase inhibitor trialled in it gave disease control more often than response,⁵ a reading composed from the response categories as the review reports them, rather than from a disease-control figure stated in either report.

The fusion junction is the one feature of an EMC tumour that exists at the RNA level and in no normal cell. An antisense gapmer tiled across it recruits RNase-H1 to cleave the transcript it pairs, and the six-nucleotide DNA gap at the centre of a 5-6-5 architecture is where that cleavage occurs. Junction-directed nucleic-acid agents are a thirty-five-year lineage, reported against six fusion oncogenes; two of the six were antisense oligonucleotides and the rest were RNA-interference agents, one of them delivered from a lentiviral vector rather than administered as an oligonucleotide.^{6,7,8,9,10,11} No such design is reported for any *NR4A3* fusion in the literature retrieved here, and that absence is why this work exists: EMC is rare enough that the design step has not been done, and rare enough that no group is likely to do it as a by-product of something else.

What a junction design must survive follows from its construction. Both halves are parent-gene sequence, so each parent matches roughly half the oligonucleotide, and half-identity falls far outside the mismatch budget of a

conventional off-target search: a parent is not returned as a near-match. RNase-H1 does not require the whole duplex, only that the gap be paired — a premise adopted here rather than established, and one whose length requirement is not the criterion this paper screens on: the enzyme is reported to need at least six paired nucleotides, with seven to ten given as the working range, while the screen below counts a liability only at ten contiguous base pairs through the gap. Whether a wild-type parent pairs the catalytic gap contiguously is therefore a separate question from overall similarity, and it is the one this work puts to all 190 designs before recommending any.

2 • The reagents

The two reagents named for synthesis are the best available designs at the two junctions with a published exon-resolved breakpoint and the highest reported prevalence: 5'-GGGCATATCATCAAAC-3' at *EWSR1* exon 12 joined to *NR4A3* exon 3, and 5'-GGGCATATCTTGTGTG-3' at *TAF15* exon 6 joined to *NR4A3* exon 3 (Table 1). Exon numbers throughout are transcript exon indices, counted from the transcript 5' end and including non-coding exons; that convention differs from coding-exon indexing for *TCF12*, *TFG* and *NR4A3*, and an acceptor exon number read under the wrong convention selects a different reagent. Both hold the panel's top gap-level margin of three, meaning three junction-unique bases inside the catalytic gap on the shorter side of the breakpoint. Neither pairs a wild-type parent through the gap at the ten-base-pair criterion adopted here. That criterion counts only windows pairing the catalytic gap in full and reaching ten contiguous base pairs, and it is the run length rather than the pairing that the cut moves: at seven base pairs the same screen returns 175 of 190 and at six, 181. A design whose gap carries a mismatch is scored zero by that screen rather than scored short, so the 87 bound the fully-paired class and not the whole parent liability, and this work reports no measurement of the mismatched class. Both reagents pair a wild-type parent through part of the gap at the *NR4A3* exon-2/exon-3 seam their acceptor halves share, neither in full; those partial duplexes are not counted here and have not been measured reagent by reagent.

Both sit close to that criterion. The *EWSR1* reagent's longest wild-type parent run is eight base pairs and the *TAF15* reagent's is nine, both against wild-type *TFG*. Where the criterion is set therefore decides how the two read: at a cut of eight both fall inside the class this work marks as not to be ordered, at nine the *TAF15* reagent alone does, and only at ten does neither. Ten is adopted as a convention and is not measured for this architecture; the extended report gives the panel's count at every cut.

One hazard of this design space belongs beside the sequences. Consecutive registers of one seam differ by a single-base slide and can carry opposite verdicts, so a design that pairs a wild-type parent through its whole catalytic gap sits one nucleotide from one that does not (Table 2). Neither member of such a pair may be substituted for the other.

Predicted transcriptome load separates the two: 123 gap-paired sense-strand near-matches for the *EWSR1* reagent at ten times the default search depth against eight for the *TAF15* one, both read from the alignment screen rather than from Table 1, which carries the parent-duplex reading and not the load. What that number is made of matters as much as its size: 82 of the 123 are predicted transcript models rather than curated records, and 32 of the remaining 41 are one gene, so the count is a search result over a database that includes predictions and not a census of expressed transcripts. The *EWSR1* reagent also carries a sense-strand near-match in wild-type *TAF15* precursor RNA at two mismatches, one of them inside the catalytic gap, spanning an intron-exon boundary. That site is the cost of the same ten shared donor bases that let one oligonucleotide span the *EWSR1*, *TAF15* and *FUS* breakpoints at once (Figure 1). The *TAF15* reagent carries no sense-strand precursor site at all. Neither load is a disqualification, and neither is a statement about safety: these are predictions from sequence search, not measured off-target activity.

Discounted by the breakpoint distribution of an 18-case series,¹² the two junctions account for 68.4% of molecularly confirmed cases in a 58-case cohort,¹³ roughly two thirds. That figure prices which published junctions the two reagents address. It is not a coverage measurement, no patient having been screened with either sequence. The range

39.9% to 82.8% quoted with it is not a confidence interval and carries no nominal coverage level: it is what the figure becomes when each breakpoint fraction is taken to its own Wilson bound while the partner shares are held at their point estimates, so two of its four inputs do not vary, and it also assumes a breakpoint distribution reported in one cohort transfers to a second collected twenty-one years later. The *TAF15* arm is priced at three of three reported breakpoints, an upper bound rather than an estimate, and reference 14 reports two major *TAF15::NR4A3* isoforms.

That figure is not a ceiling on what this panel could address. A third design, 5'-GGGCATATCTCCACGG-3' at *EWSR1* exon 13 joined to *NR4A3* exon 3, is already tiled and screened here at the same top gap-level margin, with a longest wild-type parent run of eight base pairs against wild-type *TCF12*. Its two off-target axes point in opposite directions and it should not be preferred on either: it carries the lighter transcriptome count, 24 gap-paired near-matches at two loci against 123 at six, and the heavier exposure, both of its loci being transcribed in the organs a systemic phosphorothioate gapmer distributes to where none of the exon-12 reagent's measurable loci is. It also carries a sense-strand near-match in wild-type *NR4A3* precursor RNA at two mismatches, one of them inside the catalytic gap and spanning an intron-exon boundary — the same class the *EWSR1* reagent above carries against *TAF15* — and it reaches the *NR4A3* exon-2/exon-3 seam through part of its gap, as the two named reagents do and on the same unmeasured footing. The reason to make it is coverage, which none of these axes moves. Its junction is the type-5 transcript, named in the same sentence of the 18-case series that supplies the exon-12 count, and adding it would take the figure above to 79.0% — a gain of about ten percentage points for one further oligonucleotide, with no new screen and no new retrieval. It is not named for synthesis here because no test article in §4 is established as carrying that junction: under the reported exon-2 reading of the two cell models none does, and under the exon-3 reading that §4 says also survives, USZ20-EMC1 would carry it exactly. That is a reason to settle the acceptor index before ordering rather than a property of the design.

3 • Selection from a panel of 190 designs

The two reagents above are what survived a screen applied uniformly to the whole panel. Across the 38 in-frame junctions of five modelled partners, 190 junction-spanning designs were tiled and put through five specificity screens. Of those, 87 let one of six mature wild-type parent transcripts pair their entire catalytic gap over a contiguous duplex of at least ten base pairs, and 61 of the 87 do so against wild-type *NR4A3*. That is the single largest liability class in the panel, and 85 of the 87 are paired by one of the design's own two parent genes rather than by an unrelated transcript. A second class is invisible to any screen over mature transcripts: 19 designs carry a sense-strand near-match in parent precursor RNA that pairs the gap in full and touches intronic sequence. Taken as a union rather than a sum, the two screens condemn 93 of the 190.

Lengthening the catalytic gap does not remove this liability, and the reason is arithmetic rather than empirical. Every base inside the gap comes from the donor exon or from the acceptor exon, so the junction-unique bases on the shorter side and the bases a wild-type parent can pair on the longer side tile the gap and sum to it. A longer gap therefore raises the margin available only by conceding parent-paired gap DNA at the design's own seam, and that concession is not flat: designs whose seam duplex reaches ten base pairs run from 76 of 190 to 228 of 266 to 342 of 342. The searched liability behaves differently. Across the three geometries screened — 5-6-5, 5-8-5 and 5-10-5 over the same 38 junctions — the count of designs pairing a parent at the ten-base-pair criterion is 87 of 190, 88 of 266 and 87 of 342, so the count does not fall while the share does, from 45.8% to 33.1% to 25.4%, because a longer oligonucleotide admits more junction-spanning registers per seam. At 5-10-5 the criterion is no longer independent of the geometry: a ten-nucleotide gap is itself a ten-base-pair hybrid, so every window pairing the whole gap clears the criterion by construction, and the three counts are not a reading at a constant substrate.

Three designs clear every screen applied here, two of them at any parent-duplex threshold and the third only at the ten-base-pair criterion adopted here — it pairs eight base pairs of wild-type *NR4A3* and fails at any cut of eight or less. None of the three sits at a junction any patient is reported to carry, which is what makes them mechanism controls rather than candidates. Selecting within each junction rather than across the panel is what makes the two named reagents available: 35 of the 38 junctions have a design that clears the parent screen, and all five junctions with a published exon-resolved breakpoint have one. Both of those readings are properties of the cut and not of the panel. At a criterion of nine base pairs 31 of the 38 junctions still clear and three of the five published ones do; at eight, 23 and two; at seven, 9 of the 38 clear and none of the five published junctions does. Those cuts are whole-duplex run lengths and not the enzyme's own unit, which is the DNA gap: the geometry here holds six gap nucleotides against every target, so no cut on this ladder is the enzymology restated. The availability the two named reagents rest on therefore holds at the criterion adopted here and fails at a run cut of seven, which is the sense in which the criterion is a convention rather than a measurement.

Two bounds on the cleanliness claim are load-bearing and are stated with it. The alignment screen is heuristic and censors on two ceilings: a hit list of up to 50 is requested and 15 are retained, so of the 183 filtered designs 35 are censored at the request and 101 more past the retained window, leaving 47 with hit lists short enough to assess for cleanliness at all; a count of clean designs is therefore a floor over that subset rather than a total over the panel. And search depth moves the result: most of the designs clean at the default ceiling are not clean at ten times it, as the extended report shows design by design.

The ten-base-pair criterion is adopted rather than measured, and the comparison against null models does not resolve an excess specific to this disease. Mononucleotide scrambles — the weakest of the ten nulls screened, and not the dinucleotide-preserving control §5 prescribes — meet the parent screen at 6.2% against 45.8% for the panel, but chimeras built at real exon termini of the same two transcripts, at junctions almost never reported in a patient, meet it at 40.6%. The comparison is narrower than it reads, because the panel arm is itself mostly unreported junctions: 25 of the 190 designs sit at the five junctions with a published exon-resolved breakpoint and the other 165 do not, which is the same property the chimeric null is discounted for. Most of the liability is therefore what joining two exon termini of these genes gives, and across cuts of six to thirteen base pairs the excess of the observed rate over the strongest null changes sign four times. No cut in that range is a boundary the data picks out.

4 • Test articles

Five test articles bear on the junctions this panel designs against, and they divide into two sources with opposite limits. Every design in the panel joins its donor to *NR4A3* exon 3; the two cell models are reported at exon 2, so they carry a panel junction only under the exon-3 reading their acceptor index leaves open.

Three are engineered constructs from a published functional study,¹⁴ whose exon spans that paper states verbatim. Two of them, E-N and T-N*, carry exactly the two junctions the reagents above span, so both named reagents have a stated test article. Rebuilding the constructs is the faster route and its critical path contains no laboratory that has to answer an email. What it cannot buy is biological relevance: a complementary DNA over-expressed in a heterologous background is not the disease, so such an experiment speaks to junction-selective knockdown of the intended transcript and not to activity at endogenous expression from an endogenous locus.

The other two are patient-derived, identity-clean models reported with two EMC tumours,¹⁵ USZ20-EMC1 (RRID:CVCL_C6MX) and USZ22-EMC2 (RRID:CVCL_C6MY). These are the only source of a fusion-positive EMC cell identified here. They are available on request from the originating laboratory with no repository deposit, and they are slow, at reported doubling times of five to six days as sarco-spheres passaged every two to three weeks, which constrains any exposure window. Their fusions are reported as *EWSR1* exon 13 and *TAF15* exon 6 joined to

NR4A3 exon 2 rather than exon 3, though that acceptor index is not settled: the report carries no sequenced exon-exon boundary, no transcript accession and no junction sequence, so two readings of the exon label survive, and this work's own withdrawn version arose from an error of exactly this class, so the assignment must be confirmed by sequencing before it selects a reagent. Reagents exist at both acceptors, so each line has one, and it is not in either case the same molecule as the reagent named above for the exon-3 acceptor: 5'-AGTGGGCTCTCCACGG-3' at *EWSR1* exon 13 and 5'-AGTGGGCTCTTGTGTG-3' at *TAF15* exon 6, both at the panel's top gap-level margin. Neither reaches the ten-base-pair criterion, and their longest wild-type parent runs are eight base pairs against wild-type *EWSR1* and nine against wild-type *NR4A3* respectively — the second against the acceptor parent on which §5's selectivity ratio is defined, which is a closer call than either exon-3 reagent presents. A reagent selected for one acceptor is not valid for the other.

One requirement is upstream of all of them. The breakpoint of the cell line or tumour sample used as the test article must be established at nucleotide resolution by RNA sequencing before any oligonucleotide is ordered. Every design here is specific to the exon pair it was tiled at, and none is valid for an unverified junction. Routine diagnosis does not supply the seam: break-apart *NR4A3* fluorescence in situ hybridisation detects a rearrangement irrespective of partner,³ so on its own it locates neither the partner nor the exon pair.

5 • The falsification experiment

The experimental design that would resolve the central uncertainty, an isogenic fusion-positive against fusion-negative comparison, has been published in an analogous fusion sarcoma. Fusion-specific antisense oligonucleotides against *NAB2::STAT6* in solitary fibrous tumour, evaluated against CRISPR-engineered isogenic fusion-positive and fusion-negative cells, reduced fusion expression by 58% and proliferation by 22% in vitro.¹⁶

Three assay controls are required, and a knockdown assay alone distinguishes none of them. A positive control gapmer against an abundant housekeeping transcript in the same cells separates failed delivery from a reagent that reached its target and did not cleave it, and it must carry the same 5-6-5 β-D-oxy-locked-nucleic-acid phosphorothioate geometry as the test article, since uptake and endosomal release track chemistry class; that is the chemistry of the two named reagents themselves, which the canonical sequence file carries in full. A scrambled gapmer of the same chemistry and geometry, dinucleotide-preserving so that nearest-neighbour stacking composition and the terminal bases are held as well as base composition, separates the backbone-class component of toxicity; that shuffle holds a 5' guanine run only by accident of the sequence it is drawn from — measured through the procedure used here it is retained for every draw against one named reagent and about one draw in ten against the other — so where the test article carries such a run it has to be imposed by hand. The scramble actually ordered must itself be put through the mature-parent screen before it is made, because 10.0% of dinucleotide-preserving scrambles pair a parent's whole catalytic gap at the ten-base-pair criterion and 3.9% do so against wild-type *NR4A3*. That screen is `aso_parent_gap_pairing.py`, and it takes the target window rather than the antisense strand: screening the sequence as it would be ordered searches the complement and returns a false pass, so the scramble must be reverse-complemented before it is submitted. A scramble reaching the criterion against a wild-type parent is redrawn rather than adjusted, because a single-base edit moves a design to a neighbouring register whose verdict may differ (Table 2). A fusion-negative isogenic comparator supplies the discrimination readout that neither of the other two gives, and no supplier of one is named here: such a pair has to be engineered. A fourth arm is available but not free: the three all-screen-clear designs of §3, named in the extended report, sit at *FUS* exon 8, *TAF15* exon 1 and *TCF12* exon 7, and none of the five test articles of §4 carries any of those junctions. Running that arm means engineering three further constructs, which is why it is a mechanism control rather than a fourth comparison to be had for nothing.

The decision threshold should be fixed before the experiment. What a result on it settles is the level of selectivity the top-margin reagents reach, not whether selectivity orders by margin across the panel; that is the ranking referred to below. Selectivity is the wild-type *NR4A3* half-maximal knockdown concentration divided by the fusion's, from a matched dose-response in the same wells, and the cut is 5.0, taken as a convention rather than measured for this comparison. A ratio of residual transcript at a single dose is not commensurate with it and must not be compared against the same cut: that ratio is bounded above by one divided by the fusion knockdown's complement, so at the 58% knockdown reported for the published experiment above it cannot exceed approximately 2.4 however selective the reagent is, and a cut of 5 would return falsification as arithmetic rather than as biology. The same anchor bounds the dose range the matched curves need. On a unit Hill slope a reagent whose ratio is exactly 5 puts the wild-type midpoint at a dose where the fusion is already 83.3% knocked down, and at the 58% reported for the published experiment above the wild-type transcript is only 21.6% knocked down — short of the half-maximal reading its own concentration is defined by. Potency at the level that experiment reported is therefore not sufficient for this readout: the dose range must reach far enough to resolve the wild-type midpoint, and a design that does not is uninformative rather than confirmatory.

The replicate count follows from the variance rather than being asserted. At a replicate standard deviation of 0.35 on the natural-log scale, six independent biological replicates give about 80% power to falsify a true selectivity of 3 and three give about 30%, computed from a noncentral *t*. Above a realised replicate standard deviation of about 0.65, no observed ratio at or above one can place the upper limit of a two-sided 95% interval for the ratio — built on the natural-log scale and carried back — below 5 at three replicates, so the test can fail only where the observed ratio is itself anti-selective. Such a test is void: it cannot fail on any reading that is not anti-selective, which is a different outcome from one that fails to falsify, and voidness is a property of the assay's replicate variance rather than of the design. The figure is therefore tied to the replicate count as well — about 0.65 at three replicates, 1.53 at six and 2.25 at ten — and it is not a single bar a pilot either clears or does not. The replicate count should be set from the pilot estimate with three as a floor and not a target, and the gate applied to the upper limit of a one-sided 95% chi-square bound on the pilot's standard deviation rather than to its point estimate, with that bound, its confidence level and its degrees of freedom reported, since the level moves the count the gate prescribes. A pilot whose bound lies at or above the void figure for the count proposed gives no assurance that count can falsify at this cut — the figure is defined on the realised replicate standard deviation and the bound is on the population one, so the gate is conservative rather than exact — so the decision at that point is a larger replicate count or no falsification test at all, and never three. That gate is a floor and not a sizing rule: clearing it establishes only that a run can fail, not that it is likely to when the reagent is selective. At this paper's own replicate standard deviation the smallest count the gate admits runs well under the 80% quoted above, so the count should be set for power at the selectivity to be excluded and the gate applied afterwards as a check on it. The ranking is falsified only where the upper bound of that same interval lies below the cut, never where a point estimate does; the void figures above are computed on the same construction, and a normal-approximation interval would move them. A dose range over which wild-type *NR4A3* never reaches half-maximal knockdown gives no numerator, the wild-type concentration being the numerator of the ratio, and the ratio is reportable only where the wild-type knockdown is itself resolved above a pre-stated limit of quantification — the quantity that can approach zero is the change in wild-type transcript, not its vehicle-well abundance. Both cases are uninformative on the same footing as a void test and must not be recorded as a reagent clearing the cut by default.

Two limits of that threshold belong with any result. It is defined on wild-type *NR4A3*, which is the acceptor parent; no donor parent is read by the ratio, so a reagent can clear the cut while pairing a donor transcript through its whole catalytic gap. Wild-type transcript for the reagent's own donor parent, and for the parent carrying its longest duplex, should be measured in the same wells and reported beside the ratio against no cut. No cut is stated for a donor ratio because no retrieved measurement bounds the parent case, so a threshold on it would be a number with nothing

behind it; the donor measurement is required to be reported and is not required to clear anything. A result on the stated cut can therefore falsify the ranking and cannot falsify the rationale. Multiplicity is bounded here rather than dismissed. The extended report declines a correction because the family over which an error rate would be controlled is open-ended, and that reason does not reach the design set out above, which is closed. What it contains is two ranking tests, one per named reagent at its own junction; the three all-screen-clear designs are a mechanism control and are not read against the cut, so they do not enter the family. Two tests at a nominal one-sided 2.5% carry a familywise error rate of 4.94%, and that is the rate this design runs at. A programme that also reads the two exon-2 reagents of §4 against the same cut is running four tests at 9.63%, and one that adds the control arm, five at 11.89%. A reader who does read all five against the cut is running a five-member family at 11.89%, which no longer supports the replicate counts above and needs a correction stated before the experiment rather than after it.

6 • Beyond the panel

The panel is bounded by what has been sequenced rather than by what can be designed, and the procedure that produced the 190 designs is released unchanged with the artefacts. It is not a turnkey service for an arbitrary breakpoint. The builder's published-breakpoint list is a waiver list rather than a gate: it admits seams that would otherwise be refused for a non-coding acceptor or an out-of-frame join, and every entry in it sits at a *NR4A3* exon-2 acceptor. A new seam that needs such a waiver requires the list to be extended before the procedure will emit for it, and a seam that needs none is emitted without any check that a patient has been reported to carry it. A design is certifiable where all five screens could be run on it and it cleared all five; a design one screen cannot address is uncertifiable whatever the other four return. By that definition the reagent for the third engineered construct of §4 is uncertifiable, since its seam lies outside the compartment one of the five screens covers. What the procedure yields is a candidate, not a validated reagent.

7 • Discussion

Designability is not the constraint in this disease. Junction-spanning designs exist at every in-frame *NR4A3* fusion junction modelled here, and 35 of the 38 have one clearing the parent screen at the criterion adopted here — 9 of the 38 at a run cut of seven and 6 at six, as §3 sets out, so that reading is a property of the cut. The constraint is discrimination between the fusion and its parents, and it is not resolved here. A junction design's most plausible wild-type liability is its own parent, in the mature transcript or across a splice junction in precursor RNA; both compartments are searched here before any molecule exists. A third is not: the patient's own un-rearranged *NR4A3* allele, which the extended report screens separately and which condemns designs at some acceptors this panel covers. The four reports of parental sparing cited here^{8,17,9,10} were all made on molecules already synthesised. What other groups do at the design stage is not established here, because no survey of published design pipelines was performed; the claim below that the screen-before-synthesis step is what is new is therefore a statement about this literature as retrieved and not a priority claim. The premise that sparing wild-type *NR4A3* is worth a specificity cost also deserves examination. *NR4A3* has two close paralogues and the family is functionally redundant where tested,^{18,19} which cuts against the premise. Against that, mice carrying reduced gene dosage across *NR4A1* and *NR4A3* develop mixed myelodysplastic/myeloproliferative neoplasms,²⁰ so loss of *NR4A3* is consequential rather than silent once paralogue reserve is reduced, and the family is not uniform in direction: within atherosclerosis *NR4A1* and *NR4A2* attenuate lesion formation while *NR4A3* aggravates it.²¹ The evidence is not decisive either way, and both sides of it are cited here.

Three limits bound what any test of these reagents could show. All five screens address hybridisation rather than cleavage, and none establishes that a predicted duplex forms or is cut. The method-level novelty of this work is nil,

junction-directed oligonucleotides being long established; what is new is the indication and the screen applied before synthesis. And systemic, antigen-dependent delivery of an oligonucleotide to a solid tumour remains unsolved, which is the gate this modality faces after any result reported here. Nothing in this work addresses it.

One constraint sits above all of the others and no reagent choice moves it. Every source of a test article named here ends at someone culturing cells, so the rate-limiting step is a laboratory rather than a line, a construct or an oligonucleotide.

8 • Methods

All analyses are computational and use public data. No laboratory work was performed. Full parameters, every screen's settings, the complete bounds on each claim and the per-design tables are in the extended report named under Data availability.

Canonical transcripts for the five partner genes and for *NR4A3* were obtained from Ensembl.²² Junction-spanning 16-mer gapmers were tiled in a 5-6-5 β -D-oxy-locked-nucleic-acid/DNA/ β -D-oxy-locked-nucleic-acid geometry,²³ one design per register at which the breakpoint falls inside the six-nucleotide DNA gap, which admits five per junction. That gap is the shortest length the cited source calls sufficient rather than its preferred one: it reports six nucleotides as necessary for noteworthy RNase-H activity, seven as allowing complete activity, and seven to ten as optimal for this architecture. Six is used here because the panel is built to be screened, and the genome-wide arm is not available above a 16-mer; a longer gap is a design choice this work does not test, and the three geometries compared in §3 vary the gap at fixed screening rather than proposing one. A design's gap-level margin is the count of junction-unique bases inside the gap on the shorter side of the breakpoint.

Five specificity screens were applied: an alignment screen against human RefSeq RNA, classifying each near-match by whether the catalytic gap is paired; an exhaustive transcript scan complete for substitutions within a one-mismatch budget; a screen of the parents' unspliced sequence; a mature-parent screen recording the longest contiguous duplex any of six wild-type parent transcripts forms through the catalytic gap; and a genome-wide screen over every position of GRCh38. A near-match is a transcript window pairing a design at 14 or more of its 16 positions. A design is liable where a wild-type parent pairs its whole catalytic gap over a contiguous run of ten base pairs or more, ten being adopted rather than measured. Null ensembles were built as scrambles of each design and as chimeras joining the same two parent transcripts at real exon termini, screened identically.

Tables

Every cell below is a column of *fusion-junction-aso-sequences.csv*, the canonical machine-readable record, except the test-article column of Table 1, which is a literature fact and carries its source in the caption. An oligonucleotide should be ordered from that file rather than transcribed from this page.

Table 1. The two reagents named for synthesis, with their parent-duplex label and their test article. Both are the best available design at their junction and both hold the panel's top gap-level margin. The parent-duplex column is the longest contiguous duplex any of six mature wild-type parent transcripts forms through the catalytic gap; neither reagent reaches the ten-base-pair criterion, and both sit close to it, so the length is printed rather than a pass mark. Test articles are the engineered constructs of Brenca et al. (PMID: 31020999), whose exon spans that paper states verbatim. The two patient-derived models of Bangerter et al. (PMID: 36316541) are REPORTED to carry NR4A3 exon-2 acceptors, an index the article records as unsettled, and on that reading are matched to different designs, not to these two. Nothing here has been synthesised or tested, and no sequence may be administered to any person or animal.

Table 1. The two reagents named for synthesis, with their parent-duplex label and their test article				
seam	reagent	margin	longest WT run (bp)	test article
<i>EWSR1</i> e12::NR4A3 e3	5'-GGGCATATCATCAAAC-3'	3	8 bp, wild-type <i>TFG</i>	E-N, engineered construct
<i>TAF15</i> e6::NR4A3 e3	5'-GGGCATATCTGTGTG-3'	3	9 bp, wild-type <i>TFG</i>	T-N*, engineered construct

Table 2. Four near-identical designs at two seams, two orderable and two not. Each pair is two consecutive registers of one seam differing by a single-base slide, and the two members carry opposite verdicts: the condemned member pairs its whole catalytic gap against a wild-type parent gene at the ten-base-pair criterion, and the orderable member does not. One of the condemned designs is not at an unrelated seam: 5'-CAGGGCATATCATCAA-3' is two single-base slides from 5'-GGGCATATCATCAAAC-3', a reagent this paper names for synthesis, and pairs 11 bp of wild-type *NR4A3* through its whole catalytic gap. Neither member of a pair may be substituted for the other, and neither is named for synthesis as a reagent against a patient's fusion. Δ The orderable *TCF12* design IS one of the three all-screen-clear designs the falsification experiment offers as a control arm, which is a different role at a junction no patient is reported to carry. The pairing is read from the canonical file's own cross-reference column rather than asserted here.

Table 2. Four near-identical designs at two seams, two orderable and two not				
seam	design	verdict	margin	longest WT run (bp)
<i>TCF12</i> e7::NR4A3 e3	5'-GGCATATCAAGCGCTG-3'	DO NOT ORDER	2	11 bp, wild-type <i>TCF12</i>
<i>TCF12</i> e7::NR4A3 e3	5'-GGGCATATCAAGCGCT-3'	orderable (a single-base slide)	3	8 bp, wild-type <i>NR4A3</i>
<i>EWSR1</i> e12::NR4A3 e3	5'-CAGGGCATATCATCAA-3'	DO NOT ORDER	1	11 bp, wild-type <i>NR4A3</i>
<i>EWSR1</i> e12::NR4A3 e3	5'-AGGGCATATCATCAA-3'	orderable (a single-base slide)	2	7 bp, wild-type <i>TFG</i>

Figure legends

One 16-mer spans three partners' breakpoints — and only one of the three is a junction any patient is reported to carry

The three FET-family donors are identical over the ten nucleotides before the breakpoint, and the acceptor exon is the same in all three. Every row is TARGET mRNA (sense, 5' to 3'); the reagent is the reverse complement of the shaded window.

target mRNA (sense, 5' to 3')	breakpoint	
EWSR1 e12::NR4A3 e3	A A T G G T T T G A T G A T A T G C C C T G C G	EWSR1 · reported in patients at this exon
FUS e10::NR4A3 e3	A C T G G T T T G A T G A T A T G C C C T G C G	FUS · no exon-resolved report at all
TAF15 e11::NR4A3 e3	A C T G G T T T G A T G A T A T G C C C T G C G	TAF15 · partner reported, this exon not



reagent 5'-GGGCATATCATCAAAC-3' (antisense), 8 donor and 8 acceptor bases either side of the seam

Blue, donor exon; green, NR4A3 acceptor exon; boxed and purple, the one position at which the three donors differ. Shaded box, the window this reagent targets.

The reagent is 5'-GGGCATATCATCAAAC-3', the reverse complement of the shaded window (5'-GTTTGATGATATGCC-3'). Transcribing the shaded letters orders the sense strand, which is a different molecule with no antisense activity.

Research use only, not for administration. The bases alone, ordered as unmodified DNA, are a different molecule; order from fusion-junction-aso-sequences.csv.

Figure 1. One 16-mer spans three partners' breakpoints, and only one of the three is a junction any patient is reported to carry.

The junction windows of *EWSR1* exon 12, *TAF15* exon 11 and *FUS* exon 10 joined to *NR4A3* exon 3, aligned at the breakpoint, each row carrying its own reporting status. The same paralogy that lets one reagent cover three fusions is why these designs are hard to discriminate from the parent transcripts: this reagent's gap-level margin is three junction-unique bases inside the six-nucleotide catalytic gap. Coverage is a statement about sequence, predicted rather than measured, and not a claim that one reagent serves three patient groups.

Declarations

Research use only, and not for administration to any person or animal. Every oligonucleotide sequence here is a research reagent for laboratory investigation only, and none has been synthesised or tested. Order from the canonical record, `fusion-junction-aso-sequences.csv`, rather than by copying from this article, and not at all until the breakpoint of the cell line or patient sample has been established at nucleotide resolution by RNA sequencing. That file specifies the sequence, the position of every locked residue and the backbone; it does not specify purification grade or synthesis scale, which are the orderer's to choose and are stated nowhere in this work.

Ethics approval, consent to participate and consent for publication. Not applicable. No human subjects, human material or animals were involved.

Data and code availability. All code, graded artefacts and per-design tables are public and are deposited under [doi:10.5281/zenodo.22028916](https://doi.org/10.5281/zenodo.22028916). The extended report of this work, carrying every screen's full parameters and the complete bounds on each claim, is `fusion-junction-aso-research-article.md` inside that same deposit; it is prepared for posting as a bioRxiv preprint and is not yet posted, so the archived copy is the citable one. An earlier version of these analyses placed the acceptor junction incorrectly through a coding-versus-transcript exon indexing error and was withdrawn in full; the panels were rebuilt and verified against two independent transcript acquisitions, and the complete correction record, including every superseded value, is released with the archive.

Use of artificial intelligence. A large language model (Claude, Anthropic) was used throughout this work: to write and review the analysis code, to run the screens, to retrieve and check literature, and to draft and revise this manuscript. Eleven figures in this article are pinned to the artefact that produces each and re-checked on every commit; the remaining counts are reproducible from the released artefacts but are not individually guarded, and the

decision thresholds and power figures of §5 are stated conventions and arithmetic rather than artefact readings and have no producing artefact at all. The author directed all work reported here and is responsible for its content.

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Competing interests. The author declares no financial competing interests: he holds no patent, patent application, equity or consultancy relating to any sequence or method described here. One non-financial interest belongs on the record: this work reaches a journal because its screens returned a nameable reagent, and the same procedure applied to a panel that returned nothing would be far less likely to have been written up at all, so the published record of this approach is subject to the survivorship this paper cannot correct for.

References

Numbered contiguously by order of first citation in this article. A reference cited in both this article and the extended report may therefore carry a different number in each, and a superscript here resolves only against this list. Metadata is read from retrieved bibliographic records.

1. Labelle Y, Zucman J, Stenman G, Kindblom LG, Knight J, Turc-Carel C, et al. Oncogenic conversion of a novel orphan nuclear receptor by chromosome translocation. *Human molecular genetics*. 1995;4(12):2219-2226. PMID: 8634690. doi:10.1093/hmg/4.12.2219
2. Paioli A, Stacchiotti S, Campanacci D, Palmerini E, Frezza AM, Longhi A, et al. Extraskelatal Myxoid Chondrosarcoma with Molecularly Confirmed Diagnosis: A Multicenter Retrospective Study Within the Italian Sarcoma Group. *Ann Surg Oncol*. 2021;28(2):1142-1150. PMID: 32572850. doi:10.1245/s10434-020-08737-7
3. Remiszewski P, Falkowski S, Szumera-Ciećkiewicz A, Spalek MJ, Rutkowski P, Czarnecka AM. From pathogenesis to the patient's bedside: a comprehensive review of extraskelatal myxoid chondrosarcoma. *Journal of cancer research and clinical oncology*. 2025;151(11):283. PMID: 41055792. doi:10.1007/s00432-025-06316-5
4. Stacchiotti S, Dagrada GP, Sanfilippo R, Negri T, Vittimberga I, Ferrari S, et al. Anthracycline-based chemotherapy in extraskelatal myxoid chondrosarcoma: a retrospective study. *Clinical Sarcoma Research*. 2013;3(1):16. PMID: 24345066. doi:10.1186/2045-3329-3-16
5. Stacchiotti S, Ferrari S, Redondo A, Hindi N, Palmerini E, Vaz Salgado MA, et al. Pazopanib for treatment of advanced extraskelatal myxoid chondrosarcoma: a multicentre, single-arm, phase 2 trial. *The Lancet Oncology*. 2019;20(9):1252-1262. PMID: 31331701. doi:10.1016/S1470-2045(19)30319-5
6. Skórski T, Szczylik C, Malaguarnera L, Calabretta B. Gene-targeted specific inhibition of chronic myeloid leukemia cell growth by BCR-ABL antisense oligodeoxynucleotides. *Folia histochemica et cytobiologica*. 1991;29(3):85-89. PMID: 1794439.
7. Toretsky JA, Connell Y, Neckers L, Bhat NK. Inhibition of EWS-FLI-1 fusion protein with antisense oligodeoxynucleotides. *Journal of neuro-oncology*. 1997;31(1-2):9-16. PMID: 9049825. doi:10.1023/a:1005716926800
8. Parker Kerrigan BC, Ledbetter D, Kronowitz M, Phillips L, Gumin J, Hossain A, et al. RNAi technology targeting the FGFR3-TACC3 fusion breakpoint: an opportunity for precision medicine. *Neuro-oncology advances*. 2020;2(1):vdad132. PMID: 33241214. doi:10.1093/noajnl/vdad132
9. Ward SV, Sternsdorf T, Woods NB. Targeting expression of the leukemogenic PML-RAR α fusion protein by lentiviral vector-mediated small interfering RNA results in leukemic cell differentiation and apoptosis. *Human gene therapy*. 2011;22(12):1593-1598. PMID: 21846246. doi:10.1089/hum.2011.079

10. Shao L, Tekedereli I, Wang J, Yuca E, Tsang S, Sood A, et al. Highly specific targeting of the TMPRSS2/ERG fusion gene using liposomal nanovectors. *Clinical cancer research*. 2012;18(24):6648-6657. PMID: 23052253. doi:10.1158/1078-0432.ccr-12-2715
11. Neumayer C, Ng D, Requena D, Jiang CS, Qureshi A, Vaughan R, et al. GalNAc-conjugated siRNA targeting the DNAJB1-PRKACA fusion junction in fibrolamellar hepatocellular carcinoma. *Molecular therapy*. 2024;32(1):140-151. PMID: 37980543. doi:10.1016/j.ymthe.2023.11.012
12. Panagopoulos I, Mertens F, Isaksson M, Domanski HA, Brosjö O, Heim S, et al. Molecular genetic characterization of the EWS/CHN and RBP56/CHN fusion genes in extraskelletal myxoid chondrosarcoma. *Genes, chromosomes & cancer*. 2002;35(4):340-352. PMID: 12378528. doi:10.1002/gcc.10127
13. Huang SC, Lee JC, Hsu YC, Tsai JW, Kao YC, Hsieh TH, et al. Extraskelletal Myxoid Chondrosarcomas: The Uncommon Clinicopathologic Manifestations and Significance of TAF15::NR4A3 Fusion. *Modern pathology*. 2023;36(7):100161. PMID: 36948401. doi:10.1016/j.modpat.2023.100161
14. Brenca M, Stacchiotti S, Fassetta K, Sbaraglia M, Janjusevic M, Racanelli D, et al. NR4A3 fusion proteins trigger an axon guidance switch that marks the difference between EWSR1 and TAF15 translocated extraskelletal myxoid chondrosarcomas. *J Pathol*. 2019;249(1):90-101. PMID: 31020999. doi:10.1002/path.5284
15. Bangerter JL, Harnisch KJ, Chen Y, Hagedorn C, Planas-Paz L, Pauli C. Establishment, characterization and functional testing of two novel ex vivo extraskelletal myxoid chondrosarcoma (EMC) cell models. *Human cell*. 2023;36(1):446-455. PMID: 36316541. doi:10.1007/s13577-022-00818-x
16. Li Y, Nguyen JT, Ammanamanchi M, Zhou Z, Harbut EF, Mondaza-Hernandez JL, et al. Reduction of Tumor Growth with RNA-Targeting Treatment of the NAB2-STAT6 Fusion Transcript in Solitary Fibrous Tumor Models. *Cancers*. 2023;15(12):3127. PMID: 37370737. doi:10.3390/cancers15123127
17. Lee MS, An S, Song JY, Sung M, Jung K, Chang ES, et al. Cancer-Specific Sequences in the Diagnosis and Treatment of NUT Carcinoma. *Cancer research and treatment*. 2023;55(2):452-467. PMID: 36265509. doi:10.4143/crt.2022.910
18. Freire PR, Conneely OM. NR4A1 and NR4A3 restrict HSC proliferation via reciprocal regulation of C/EBPα and inflammatory signaling. *Blood*. 2018;131(10):1081-1093. PMID: 29343483. doi:10.1182/blood-2017-07-795757
19. Beard JA, Tenga A, Chen T. The interplay of NR4A receptors and the oncogene-tumor suppressor networks in cancer. *Cellular signalling*. 2015;27(2):257-266. PMID: 25446259. doi:10.1016/j.cellsig.2014.11.009
20. Ramirez-Herrick AM, Mullican SE, Sheehan AM, Conneely OM. Reduced NR4A gene dosage leads to mixed myelodysplastic/myeloproliferative neoplasms in mice. *Blood*. 2011;117(9):2681-2690. PMID: 21205929. doi:10.1182/blood-2010-02-267906
21. Hamers AA, Hanna RN, Nowyhed H, Hedrick CC, de Vries CJ. NR4A nuclear receptors in immunity and atherosclerosis. *Current opinion in lipidology*. 2013;24(5):381-385. PMID: 24005216. doi:10.1097/mol.0b013e3283643eac
22. Dyer SC, Austine-Orimoloye O, Azov AG, et al. Ensembl 2025. *Nucleic acids research*. 2025;53(D1):D948-D957. PMID: 39656687. doi:10.1093/nar/gkae1071
23. Kauppinen S, Vester B, Wengel J. Locked nucleic acid (LNA): High affinity targeting of RNA for diagnostics and therapeutics. *Drug discovery today. Technologies*. 2005;2(3):287-290. PMID: 24981949. doi:10.1016/j.ddtec.2005.08.012